

GPU Hardware Inspection Report

Numba CUDA Device Diagnostics Summary

1. Overview

This report provides a technical summary of the detected NVIDIA graphics processing unit discovered via the Numba CUDA runtime environment. The target hardware has been verified as fully supported for high-performance parallel computing and GPU-accelerated workloads.

2. Device Specification Summary

Hardware Attribute	Specification Value
Device Name	Tesla T4
Compute Capability	7.5 (Turing Architecture)
PCI Device ID	4
PCI Bus ID	0
UUID	GPU-0a72d1e4-6aeb-6cc1-cc3a-3560dcc39d9e
Watchdog Status	Disabled
FP32 / FP64 Performance Ratio	32

3. Core Architecture & Memory Metrics

Metric Category	Recorded Data
Multiprocessor (SM) Count	40 Streaming Multiprocessors
CUDA Core Count (Total)	2,560 FP32 Cores (64 Cores per SM)
Total Device Memory	14.56 GB
Currently Used Memory	0.10 GB
Available Free Memory	14.46 GB

4. Diagnostic Notes

The diagnostic script successfully initialized the context on the target device. Memory allocation indicates optimal idle status with negligible baseline utilization (approx. 0.10 GB). The Tesla T4 is fully prepared for executing custom CUDA kernels, deep learning inference, or numeric simulation workloads.

5. Code Template

The diagnostic inspection was performed using the following Python script with Numba CUDA:

```
from numba import cuda

gpus = cuda.gpus.lst

for gpu in gpus:
    with gpu:
        device_name = gpu.name.decode('utf-8') \
            if isinstance(gpu.name, bytes) else gpu.name
        print("Device:", device_name)

        current_device_selected = cuda.detect()
        print("Multiprocessor Count:", current_device_selected)

        ctx = cuda.current_context()
        free_mem, total_mem = ctx.get_memory_info()

        free_gb = free_mem / (1024 ** 3)
        total_gb = total_mem / (1024 ** 3)
        used_gb = total_gb - free_gb

        print("\n")
        print("Extra Info:")
        print(f"Total Memory: ({total_gb:.2f} GB)")
        print(f"Used Memory: ({used_gb:.2f} GB)")
        print(f"Free Memory: ({free_gb:.2f} GB)")
```